

PC POWER MANAGER

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Very often we forget to switch off the connected peripherals like monitor, scanner and printer while switching off our PC. This leads to needless energy consumption and possible shortening of the life of the peripheral. PCs with an ATX switch-mode power supply (SMPs) unit are not provided with a mains switch outlet. It is therefore not possible to achieve automatic switching (on/off) of peripheral units with the computer power switch.

Here is a simple circuit that turns the connected peripherals on/off along with your PC. It consists of a regulated power supply, a simple USB interface and two electromagnetic relays used as power switches.

The power supply for the circuit is derived from the AC mains via transformer X1. The 15V AC available at the secondary winding of transformer X1 is first rectified by a bridge recti-

fier comprising diodes D1 through D4, smoothed by capacitors C1 and C2, and regulated by IC LM7812 (IC1). The regulated 12V DC is used to energise relay RL1. LED1 works as a power-'active' indicator.

To set up the circuit, first connect the input socket (SOC1) of the circuit to a proper AC mains wall outlet using a three-core power cable. Now connect one end of a standard USB cable to the B-type USB input socket and the other end of the cable to any vacant USB port (A-type) of the PC. Finally, plug one standard four-way switchboard (extension cord) into the supply output socket (SOC2) of the circuit and take power from this switchboard to activate all loads like monitor, scanner, printer and even your PC.



To activate the PC manager circuit, proceed as follows: Press 'start' switch S1 and hold it in this position for a few minutes. When power-'active' indicator LED1 lights up, relay RL1 energises and the 230V mains power supply from SOC1 is fed to output socket SOC2 through the contacts of relay RL1.

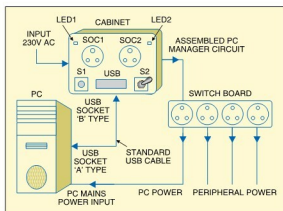


Fig. 2: Wiring diagram for PC power manager

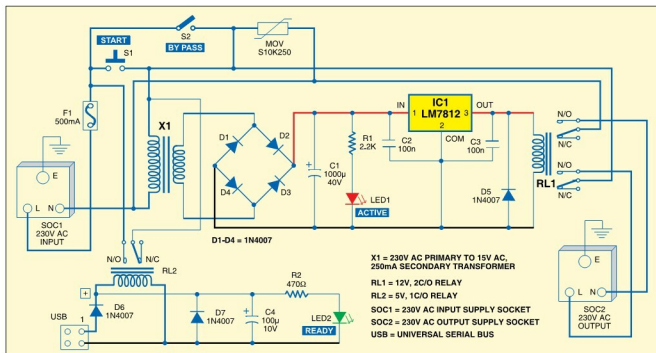


Fig. 1: Circuit of PC power manager